

NYKAA PRODUCT ANALYSIS USING WEB SCRAPING, DATA CLEANING & EXPLORATORY DATA ANALYSIS (EDA)

7 Abstract

The rapid growth of e-commerce has generated a vast amount of product data that can be analyzed to understand customer preferences, pricing strategies, discounts, and product performance. This project focuses on collecting product information from the Nykaa website using web scraping techniques, cleaning and analyzing the data, and creating an interactive Visualizations.

The dataset includes details such as product names, brands, categories, prices, discounts, ratings, and customer reviews. After preprocessing the data, exploratory data analysis (EDA) was performed to identify trends and patterns. Finally, an interactive Visualizations was developed to visualize key insights, enabling better decision-making and product analysis.

7 Introduction

Nykaa is one of India's leading beauty and personal care e-commerce platforms. It offers thousands of products from various brands across multiple categories. Since product information changes frequently, web scraping provides an efficient way to collect up-to-date product data for analysis.

This project demonstrates the complete data analytics workflow:

1. Web scraping
2. Data cleaning
3. Exploratory Data Analysis (EDA)

The project helps understand pricing patterns, product distribution, discounts, customer ratings, and brand performance.

7 Business Understanding

Nykaa is one of India's leading beauty and personal care e-commerce platforms. Understanding customer preferences, pricing strategies, discounts, and product ratings helps businesses improve product positioning, optimize pricing, and enhance customer satisfaction. This project analyzes Nykaa product data to identify trends that support better business decisions.

7 Problem Statement

Online shopping platforms contain a large amount of product information, making it difficult to manually analyze market trends. Businesses and analysts need an automated approach to collect, process, and visualize this data.

The objective of this project is to scrape product data from Nykaa, analyze it, and build an interactive dashboard that provides meaningful business insights.

7 Objectives

- Collect product information from the Nykaa website using web scraping.
- Store the scraped data in a structured dataset.
- Clean and preprocess the collected data.
- Perform exploratory data analysis.
- Analyze pricing, ratings, brands, and product categories.
- Create an interactive visualizations.
- Generate business insights to support decision-making.

7 Tools and Technologies

Tool	<u>Purpose</u>
Python	Web scraping and data analysis
Beautiful Soup	HTML parsing
Requests	Accessing web pages
Pandas	Data cleaning and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Selenium	Web scraping
Excel/CSV	Data storage
Statistical	Data Visualization

⑦ Methodology

Phase 1: Web Scraping

- Access the Nyka product pages.
- Extract product details.
- Store the extracted information in CSV format.

Data Collected

- Product Name
- Brand
- Category
- Price
- Discount
- Rating
- Website: Nykaa
- Library Used: Requests and Beautiful Soup
- Total Rows Scraped: 140
- Total Columns: 7
- File Format: CSV

Phase 2: Data Cleaning

The collected data was cleaned using Python and Pandas.

Cleaning included:

- Removing duplicate records
- Handling missing values
- Correcting data types
- Standardizing column names
- Removing unwanted symbols
- Converting prices into numeric values

Phase 3: Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset.

Analysis included:

- Product distribution by category
- Brand-wise product count
- Price analysis
- Rating analysis
- Discount analysis

Phase 4: Dashboard Development

Matplotlib was used to create an interactive dashboard.

The dashboard provides:

- Plots
- Interactive analysis

7 Dataset Description

The dataset consists of product information collected from the Nykaa website.

Example fields include:

Column	Description
• Product Name	Name of the product
• Category	Product category
• Price	Selling price
• Discount	Discount percentage
• Rating	Customer rating

7 Exploratory Data Analysis

The analysis focused on identifying:

- Top 10 expensive products

- Top 10 discounted products
- Most Popular product
- Average product prices
- Price distribution

7 Visualizations

- Products by Category
- Original Price Distribution
- Selling price Distribution
- Selling price vs No of Products Histplot
- Average Selling price by Category
- SellingPrice vs Rating Scatter Plot

7 Key Insights

- A few products are much more expensive than the others.
- Products with both low and high selling prices receive similar ratings.
- Healthcare products have the highest average selling price among all categories.
- Baby Care products have the lowest average selling price.
- Beauty products have an average selling price between Healthcare and Baby Care.
- The number of products decreases as the selling price increases.
- Beauty has the highest number of products compared to the other two categories.
- Healthcare and Baby Care have a similar number of products in the dataset.

7 Advantages

- Automated data collection

- Reduced manual effort
- Easy visualization of product data
- Interactive reporting
- Better understanding of market trends
- Supports business decision-making

⑦ Limitations

- Website structure changes may affect scraping scripts.
- Data reflects only the scraping period.
- Some product information may be unavailable or missing.
- Dynamic content may require additional scraping techniques.

⑦ Future Scope

- Integrate live data updates.
- Include sentiment analysis of customer reviews.
- Compare products across multiple e-commerce platforms.

⑦ References

- Nyka Official Website
- Python Documentation
- Pandas Documentation
- Beautiful Soup Documentation

⑦ Conclusion

This project analyzed 140 Nykaa products from the Beauty, Healthcare, and Baby Care categories using data cleaning, exploratory data analysis (EDA), and visualization.

1. Most Nykaa products are affordable, with prices below ₹1,000.
2. Beauty has the highest number of products, while Healthcare products have the highest average price.

3. Most products have good ratings (4–5 stars), showing high customer satisfaction.
4. Only a few products are premium-priced, while most are budget-friendly.
5. Product ratings do not depend on the selling price, as both low- and high-priced products receive similar ratings.
6. Overall, Nykaa offers a wide range of quality products across Beauty, Healthcare, and Baby Care categories.